

Climate Variability and Inter-State Migration in India

Ingrid Dallmann* and Katrin Millock†

*RITM, Univ. Paris-Sud, Université Paris-Saclay, Sceaux, France. e-mail: ingrid.dallmann-gamarra@u-psud.fr and †Paris School of Economics, CNRS, Paris, France. e-mail: millock@univ-paris1.fr

Abstract

We match climate data to migration data from the 1991 and 2001 Indian Censuses to investigate the impact of climate variability on internal migration. The article makes four contributions to the existing literature on macro-level migration flows. First, use of census data allows us to test and compare the effect on migration of climatic factors prior to migration. Second, we introduce relevant meteorological indicators of climate variability, to measure the frequency, duration, and magnitude of drought and excess precipitation based on the Standardized Precipitation Index. Third, we estimate the total effect (direct and indirect effects) of climate variability on bilateral migration rates. Fourth, we examine three possible channels through which climate variability might induce migration: average income, agriculture, and urbanization. The estimation results show that drought frequency in the origin state increases inter-state migration in India. **This effect is stronger in agricultural states, and in such states the magnitude of drought also increases inter-state migration significantly. Drought frequency has the strongest effect on rural–rural inter-state migration.** (JEL codes: O15, Q54).

Key words: climate variability, drought, excess precipitation, India, internal migration, PPML, SPI

1. Introduction

Negative effects linked to climate variability are becoming ever more apparent. It is causing both increased numbers of natural disasters resulting in huge economic and human losses, and long-term consequences on the economy and on population distribution. The most recent Intergovernmental Panel on Climate Change (IPCC) assessment report discusses the different ways that climate may affect migration, although expected flows are difficult to quantify (IPCC, 2014). The detailed studies in a report commissioned by the UK government (Government Office for Science, 2011) show that environmental change will affect migration in the present and in the future but that its influence will be evident principally in its economic, social, and political effects. Climate variability can have particular direct effects, such as degraded health, increased mortality risk, capital destruction and disruption

to socioeconomic activities, and also indirect effects—on the environment and the economy—through price and wage adjustments in the market, which directly or indirectly induce migration. The objective of this article is to test the hypothesis that climate variability acts as a push factor that increases internal migration in India.

We match bilateral migration data from the 1991 and 2001 Indian Census with state-level climate data. We estimate bilateral migration rates to control for important existing migration determinants in origin and destination states, and account for zero flows using a Poisson pseudo-maximum likelihood (PPML) estimator. The article makes four contributions to the existing literature on macro-level migration flows. First, the advantage of the current study is that it uses the 1-year migration definition from the Indian Census which enables an exact timing of climatic factors prior to migration and the observed migration flow which in turn allows us to rule out simultaneity. Existing studies rely on average migration flows over 5- or 10-year periods linked to average climate anomalies over the same 5- or 10-year periods. We compare estimations based on 10-year averaged migration flows on 10-year average climate variability with estimations based on a 1-year migration flow and climate variability before migration. Second, we introduce relevant meteorological indicators of climate variability based on the Standardized Precipitation Index (SPI). The SPI measures anomalies in rainfall compared to the long-run average defined from 1901 up to the year of the census. The advantage provided by the SPI versus other measures used in the literature is that it allows comparability across states, and the possibility to measure not only the magnitude but also the frequency and duration of droughts and excess precipitation. Third, unlike most existing studies, we estimate the total effect of climate variability on bilateral migration rates, addressing possible over-controlling bias by excluding income and other migration determinants dependent on climate from the estimations of the total effect of climate variability. Fourth, we examine separately three possible channels through which climate variability might induce migration: total income, agriculture, and urbanization.

The estimation results show that drought frequency has a significant impact on inter-state bilateral migration rates when controlling for migration costs, origin-state characteristics, and destination-state pull factors. Each additional month of drought in the origin state during the 5 years preceding the year of migration increases the bilateral migration rate by 1.5% averaged over all states, and by 1.7% for agricultural states. In the case of agricultural states in particular, the magnitude of the drought also has an effect. The results are robust to controlling for the area of irrigated land in the state, and the inclusion of controls for all time-invariant bilateral fixed effects. The relative effect of climate variability is quite small compared to the effect of migration costs measured by the barriers to inter-state migration. When exploring several potential channels for the indirect effect of climate variability on bilateral inter-state migration rates in India, we find evidence that part of the mechanism works through total income and agricultural income. Inter-state migration is driven not just by agricultural income but by total income in the destination state compared to the origin state. Controlling for the urbanization rate in the state of origin does not change the effect of drought frequency on bilateral inter-state migration rates, and we conclude that the effect does not work via urbanization in the case of Indian inter-state migration. A novelty of our study compared to the literature is that we analyze actual migration flows across rural and urban areas at the inter-state level, and thus can test the effect of climate variability on these flows as well as on state aggregate urbanization rates. Decomposition of total inter-state migration into rural–rural and rural–urban migration

shows that the effect of drought frequency is the strongest on rural–rural inter-state migration rates.

We find also that a higher number of months with excess precipitation lower bilateral inter-state migration, which is contrary to the expected *ex ante* effect if excess precipitation measures flood events. This means that either floods need to be represented by different measures than excess precipitation or the migration response is different after floods. Following additional tests with alternative flood indicators, we argue that the results indicate that excess precipitation or floods matter less than droughts for explaining permanent inter-state migration in India.

The article contributes to a growing literature that analyzes the link between migration and climate variability. The idea that negative environmental conditions increase international migration was proposed in the ‘environmental refugees’ literature (Myers, 1997) but was re-interpreted and moderated by Piguet (2010) and Gemenne (2011) among others. Several studies use detailed microeconomic data to analyze the factors linking migration to climatic conditions. For example, in a large household study of Bangladesh, Gray and Mueller (2012) found that floods had no significant impact on migration but that weather-related crop failures increased migration. In another study that relates counts of natural disasters to permanent migration inferred from the Indonesian Family Life Surveys data, Bohra-Mishra et al. (2014) find no significant impact of natural disasters other than landslides on internal migration of entire households but find a significant and large effect of temperature and a significant but smaller effect of rainfall. This stream of the literature, which is reviewed in Lilleor and Van den Broeck (2011), shows how individual household factors contribute to vulnerability and explains what makes some households migrate and others not. However, it is difficult to generalize the findings from these studies to other countries.

Macroeconomic studies on international migration flows, such as Reuveny and Moore (2009), Beine and Parsons (2015), Coniglio and Pesce (2015), Cattaneo and Peri (2016), and Cai et al. (2016), test for the effects on cross-border flows. Reuveny and Moore (2009) show that both weather-related natural disasters and climate anomalies may (directly) induce increased migration into the Organisation for Economic Cooperation and Development (OECD) countries. In a comprehensive study of international migration over the period 1960–2000, Beine and Parsons (2015) find no effect of either temperature or rainfall deviations on international bilateral migration flows, including south-south migration which is an important difference compared to OECD migration data. Coniglio and Pesce (2015) test additional definitions of weather variables and find evidence of a positive effect of rainfall inter-annual variability on out-migration to OECD countries. These different findings are due in part to the use of different data sets: Beine and Parsons (2015) use migration flows calculated from migration stock data at 10-year intervals from 1960 to 2000, while Coniglio and Pesce (2015) and Cai et al. (2016) use annual data over a shorter time span (1990–2001 and 1980–2010). Cattaneo and Peri (2016) analyze the heterogeneous response in relation to income levels. Using the same data as Beine and Parsons (2015), they find that higher temperatures increase out-migration in middle-income countries, whereas in poor countries higher temperatures reduce out-migration.

The current article adds to a recent strand of work analyzing climatic factors and migration that relies on the most comprehensive data on migration flows at a country level, that is census data. Few studies use census data to study climatic factors and internal migration in large countries, and those that do focus mainly on the USA (Boustan et al. 2012; Feng

et al. 2012).¹ Feng et al. (2012) study the indirect effect of temperature-induced crop shocks on out-migration from the US corn belt states, while Boustan et al. (2012) show that floods and tornadoes had a significant effect on gross migration flows in the USA in the 1920s and 1930s. Our study is the first to use census data to analyze bilateral internal migration rates in a large developing economy such as India.

We focus on internal (inter-state) migration in India where migration induced by climate variability is more likely to occur within national borders due to migration costs and legal barriers (Marchiori et al. 2012; Beine and Parsons 2015). Also, low-income and lower-middle-income countries are more vulnerable to climate variability than high-income countries (Stern 2007; Government Office for Science 2011) due to their ability to adapt and their geographical location. To account fully for all possible factors influencing migration, we need to study bilateral flows which prohibit use of the more detailed district-level data in the Indian Census, which record the destinations but not the origins of migrants. We contribute to the migration literature which typically uses gravity-type models that incorporate socioeconomic but not environmental factors (Karemera et al. 2000; Mayda 2010; Van Lottum and Marks 2010; see in particular Özden and Sewadeh 2010, for India).

The only other studies of migration and climate in India analyze either cross-section household-level data from the National Sample Survey (NSS), as in Kumar and Viswanathan (2013), or use census data to apply Feng et al.'s (2012) method to study migration induced by agricultural shocks (Viswanathan and Kumar 2015). The state-level analysis in Viswanathan and Kumar (2015) shows that weather-induced shocks to agricultural income induce out-migration for employment reasons. The objective of our analysis is to measure the total effect (direct and indirect effects) on internal migration. Also, our study uses complete census data (31 of the 32 states according to the 1991 state borders) for 1991 and 2001, while Viswanathan and Kumar (2015) analyze data for 15 major states over the period 1981–2001. In addition, they analyze out-migration rates at state level and in-migration rates at district level, while we analyze bilateral migration rates.

The remainder of the article is organized as follows. Section 2 presents the context and statistics for climate variability and inter-state migration in India. Sections 3 and 4 describe the empirical estimation strategy and the data. Section 5 presents the empirical results, and Section 6 concludes.

2. Inter-State Migration and Climate Variability in India

Analyzing inter-state migration in India is particularly appropriate for a study of internal migration because of the heterogeneity among states in relation, especially to demography and climate. Measured by the Environmental Vulnerability Index,² India is considered extremely vulnerable because of both its climate and its population density. India has a large range of climatic regions from tropical in the South to temperate and alpine in the Himalayan North. The main natural disasters in India are drought, flood, and tropical cyclones, measured by the number of people affected (Attri and Tyagi, 2010). In the present analysis, we focus on droughts and excess precipitation. India is the second most populous country in the world with 1210 million inhabitants in 2011 which represents 17.5% of the

1 Mastorillo et al. (2016) analyze South Africa.

2 Index developed by the South Pacific Applied Geoscience Commission (SOPAC) and the United Nations Environment Program (UNEP).

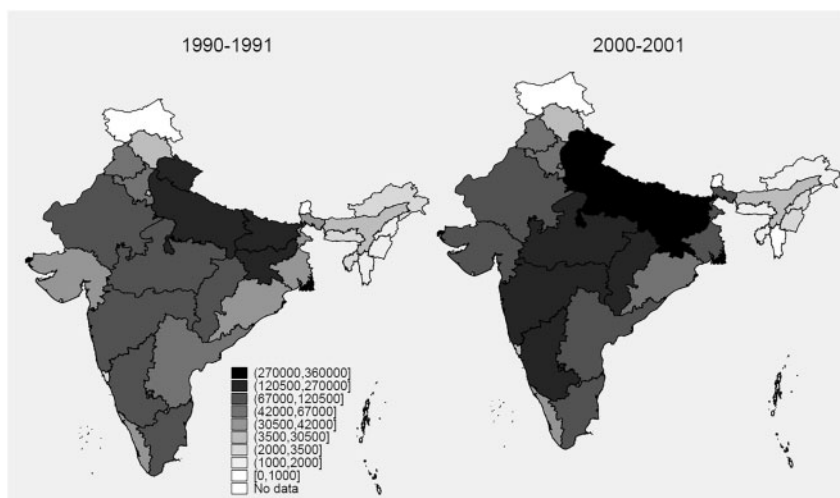


Figure 1. Maps of Indian inter-state out-migration in 1991 and 2001, by state.

Note: Migrants are defined as individuals declaring the last place of residence in year $t - 1$ as being different from the place of residence in year t declared in the census.

Source: Authors' calculations based on 1991 and 2001 Indian Census, D2-Series.

world population on only 2.4% of the world surface area, and a population growth between 2001 and 2011 of 17.6%. Its population is mainly rural—69% in 2011 or 833.5 million people (Census of India 2011). Population densities differ widely among states ranging from 17 to 11,297 persons/ km² in 2011 (Arunachal Pradesh and Delhi, respectively). In 1991, 26.7% of the total population was internal migrants including 11.8% inter-state migrants. In 2001, these figures increased to 30.1% (310 million persons) and 13.4%. International migration is only 3.8% in India, according to the 64th round of the NSS conducted in 2007–2008 (Czaika 2011). These statistics suggest potential influence of climate variability on internal migration.

Figure 1 shows the number of out-migrants by state in 1990–1991 and 2000–2001. It confirms Özden and Sewadeh's (2010) finding of the major northwestern migration corridors based on data from the 55th round of the NSS in 1999–2000. The states with the highest numbers of inter-state out-migrants are the northern states Uttar Pradesh and Bihar, the central state Madhya Pradesh, and the southwestern states Maharashtra and Karnataka (darker shades).

Figure 2 shows the average SPI for the 5 years preceding the migration flows (1986–1990 and 1996–2000) for illustrative purposes. It ranges from -1 to $+1$, which represents moderate deviations. The lighter shades indicate negative values, and thus a precipitation deficit compared to the long-run mean; the darker shades indicate excess precipitation. Comparison of Figures 1 and 2 shows that before 1991 the major out-migration states all had negative SPI values on average. In the southwestern states of Karnataka and Maharashtra, the average SPI returned to around zero in 2001, while in Bihar (in the North) the average SPI became more negative.

In the econometric analysis, one of the main measures of climate variability that we will use is the frequency of months when the SPI was at least 1 standard deviation above or

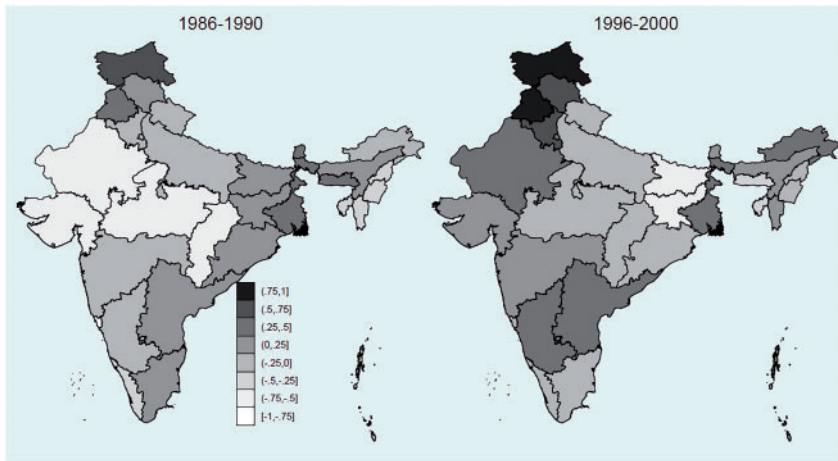


Figure 2. Maps of Indian average SPI by state, 1991 and 2001.

Source: Authors' calculations based on the CRU TS3.21 data.

below its long-run mean. Figure 3 shows the variability in the measure between the two censuses. It shows the number of months with one standard deviation or more of either low precipitation ('drought') or excess precipitation ('flood') in the 5 years preceding the censuses in 1991 and 2001. The first thing to note is that the months with drought by state varied widely between 1991 and 2001, but there is less variation over time in the number of months with excess precipitation by state. The decade 1981–1990 was a dry period in India (Attri and Tyagi 2010). Overall, several of the states recorded no occurrences of drought or excess precipitation at all in the 5 years preceding 2001. The states with a high number of months with low precipitation in the 5 years preceding 1991 are Kerala and Madhya Pradesh and several small states and island states, and in 2001 they are Bihar, Tripura, and Nagaland. The states with the most months of excess precipitation in the 5 years preceding 1991 are Himachal Pradesh, Haryana, Meghalaya, Punjab, Chandigarh, and Andhra Pradesh, and in the 5 years preceding 2001 are Haryana, Jammu and Kashmir, Rajasthan, Himachal Pradesh, and Punjab.

Comparison of the frequency of drought and excess precipitation frequencies with the migration data shows that the four states with the highest out-migration in the years studied (Uttar Pradesh, Bihar, Madhya Pradesh, and Maharashtra) all experienced drought episodes, and especially the major out-migration states of Bihar and Madhya Pradesh. These states all experienced less than 12 months of excess precipitation in the 5 years preceding the 1991 census, and no periods of excess precipitation in the 5 years preceding the 2001 census.

3. Empirical Specification and Method

3.1 Theoretical framework and econometric specification

We base the econometric specification on a random utility model (as in Beine et al. 2011; and Beine and Parsons 2015), where people can choose to stay in their state of residence or

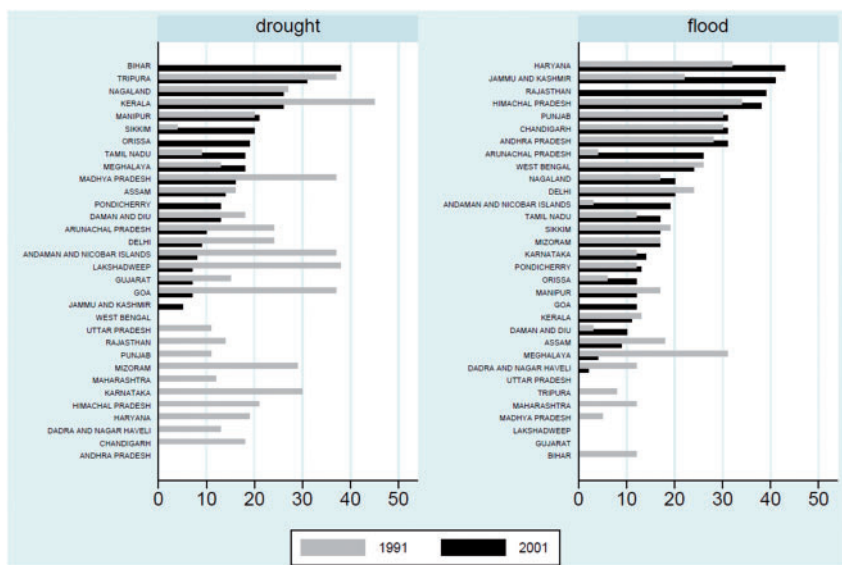


Figure 3. Frequency of low precipitation ('drought') and excess precipitation ('flood') by state, 1991 and 2001.

Note: Frequency of low and excess precipitation is defined as the number of months when the standardized precipitation index (SPI) was at least 1 standard deviation below/above its long-run mean.

Source: Authors' calculations based on the CRU TS3.21 data.

to migrate to another state to maximize their utility. The model specifies the determinants of bilateral migration and controls for factors 'pushing' and 'pulling' migration:³

$$\ln \frac{m_{ij,t}}{m_{ii,t}} = \ln \frac{w_{j,t}}{w_{i,t}} + S_{j,t} - S_{i,t} - C_{ij,t}, \quad (1)$$

where $m_{ij,t}$ is the bilateral migration rate from state i to state j during year $t-1$ to t , and $m_{ii,t}$ is the stock of the initial population staying in state i during year $t-1$ to t . The determinants are the log of the ratio of the per capita income in the destination state ($w_{j,t}$) and the per capita income in the origin state ($w_{i,t}$), the destination state characteristics ($S_{j,t}$), the origin-state characteristics ($S_{i,t}$), and the cost of migration from state i to state j at time t ($C_{ij,t}$).

Since income is endogenous to climate (Dell et al. 2009; Burke et al. 2015), we do not include it in our base estimation. Instead, we follow Dell et al. (2014) and exclude the income ratio, since it would bias the measure of the total effect of climate variability on migration, given that climate is a determinant of income. This results in the following econometric specification:

$$\ln \frac{m_{ij,t}}{pop_{ii,t}} = \alpha_0 + \alpha_1 \sum_{t=5}^t clim_{i,t} + \alpha_2 \ln(SC_{i,t} + 1) + \alpha_3 \ln(ST_{i,t} + 1) + \alpha_4 M_{ij,t-5} + \alpha_5 d_{ij} + \alpha_6 b_{ij} + \alpha_7 l_{ij} + \gamma_i + \delta_{j,t} + \epsilon_{ij,t} \quad (2)$$

3 For details of the model, see Beine and Parsons (2015).

The principal variables of interest are those for climate variability ($clim_{i,t}$). We hypothesize that precipitation variation is a push factor in migration. This applies to the case of developing countries, where poor people move not as a result of comparing origin and destination climatic factors but to escape drought or floods which affect their well-being. Accordingly, our variability and adverse weather events variables apply only to the origin state. We define drought and excess precipitation events, based on the SPI, and differentiate between frequency, magnitude, and duration of events during the 5 years preceding migration.

Origin-state characteristics, $S_{i,t}$, include time-varying and time-invariant factors. We include scheduled castes (SCs) and scheduled tribes (STs) rates as a percentage of the total population in the state of origin. In India in 2001, 16.2% of the population belonged to a SC known as 'the untouchables', and 8.2% belonged to STs. Most work on Indian migration takes account of these two factors to examine the role of social factors in the migration decision (Bhattacharya 2002; Mitra and Murayama 2008). The Hindu varna system classifies Indian society into groups based on caste, ethnicity, and religion. This classification is reflected in the labor force participation (Dubey et al. 2006). Iversen et al. (2014) show that SCs do better in villages where they are the majority, which may make them less likely to move.⁴ Indeed, Bhattacharya (2002) finds that SC incidence in rural areas is associated with lower out-migration rates, whereas the percentage of STs has no statistically significant effect. Hnatkovska and Lahiri (2015), using the NSS over the period 1983–2008, show that migrants are less likely to be members of backward castes as measured by the proportion of SC/ST. This also goes in line with Munshi and Rosenzweig (2016) who argue that one of the main reasons for India's low urbanization rate is the benefit of caste-based insurance networks in the rural origin villages.

We include time-invariant origin-state fixed effects (FEs) (γ_i) to capture the vulnerability of the geographic zone, in particular mountains, low-level coastal areas, and arid lands. This dummy controls also for the states affected by the 1958 Armed Forces (Special Powers) Act. This Act gives special powers to the armed forces (military and air forces) in so-called 'disturbed' areas of Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, and Tripura. These states have experienced violence which may have induced migration. Migration varies also depending on the employment opportunities in the destination state's labor market, and the education opportunities. These time-varying characteristics of the destination state ($S_{j,t}$) are captured by destination-time fixed effects ($\delta_{j,t}$) including any potential climate pull effect.

The costs of migration ($C_{ij,t}$) are represented by migration networks ($M_{ij,t-s}$), distance (d_{ij}), and dummy variables for common border (b_{ij}) and language (l_{ij}) between states.⁵ Migrant networks are time-variant and affect migration by reducing information and assimilation costs, among others. The networks are measured as the stock of past migrants

4 If they migrate, they are more likely to choose destinations where there are members of their own subcaste. Since we do not have data on subcastes (jati), we cannot construct an appropriate measure of caste networks. However, we can control for the lower probability of migration of SCs and STs by the rate of SC/ST in the origin state.

5 Distance, common border, and common language are frequently used in bilateral migration analyses to measure the monetary and nonmonetary costs of migration (Bodvarsson and Van den Berg, 2009). Migrant networks are also important determinants of migration (Munshi, 2003; Beine et al., 2011).

from the same state i residing in the state j , as a percentage of the total population in state j , at time $t-5$. We define the network at $t-5$ to avoid simultaneity of the migration flows, present in the network definition, with the dependent variable and the climatic variables. All control variables are defined in detail in [Supplementary Appendix A](#).

The expected signs on the variables representing the costs of migration are: $\alpha_4 > 0$, $\alpha_5 < 0$, $\alpha_6 > 0$, and $\alpha_7 > 0$. Migrant networks are expected to have a positive effect on bilateral migration rates. The relation between migration and distance is negative and proxies for migration travel costs. A common border and a common language reduce the cost of migration and proxy for cultural similarities between states. As argued above, all else being equal, the rate of SCs should imply lower migration, and we expect $\alpha_2 < 0$. We would expect a similar effect for STs, although STs live in states with more events of violence which could increase out-migration. Ex ante, the coefficient of α_3 could be positive or negative.

For the variables representing climate variability, we expect a positive sign ($\alpha_1 > 0$) for the drought measures which should act to push migration. The sign of excess precipitation is uncertain ex ante because excess precipitation could be a proxy for climatological floods and then $\alpha_1 > 0$. However, flood events depend also on the topology of the land and the geography (rivers; coast) as well as precipitation. Excess precipitation can be associated to better quality of land and growing conditions, and hence, one might expect $\alpha_1 < 0$ for excess precipitation that is at a lower level than extreme precipitation.

3.2 Estimation method and econometric issues

We start by discussing to what extent a causal interpretation of the results of [Equation \(2\)](#) can be inferred. We use exogenous weather data to construct the climate variability variables. This should reduce several potential sources of violation of the zero conditional mean assumption. However, three important econometric issues arise from [Equation \(2\)](#). First, we do not include the income ratio of Model (1) in [Equation \(2\)](#). [Angrist and Pischke \(2009\)](#) discuss the implications of using inappropriate controls, that is controls that are also outcome variables (Section 3.2.3). They explain why it is better to exclude such variables, even if they are correlated with the independent and the dependent variables. If the objective is to measure the causal effect of climate variability on migration, adding income to the regression introduces a bad control and biases the coefficient of climate variability. Indeed, according to [Dell et al. \(2009\)](#) and [Burke et al. \(2015\)](#), among others, income is endogenous to climate.⁶ Good control variables for our research question are variables that were fixed at the moment of the climate variability event. If we include income, what we are measuring is the effect of the climate variability for a given income, which will result in selection bias. We thus measure the total effect of climate variability, holding other origin-state characteristics invariant in time, and destination state characteristics and cost of migration fixed. Controlling for origin-state characteristics is important, since they may be highly correlated to climate and migration. Nevertheless, potential omitted variables bias can still arise if the correlated climate variables are not included. We take this into account in the robustness checks performed in Section 5.3.

6 As expected, we find also that the income ratio is significantly affected by the climatic factors (see [Supplementary Appendix Table SD3](#)).

Second, in relation to the functional form, the specification in Equation (2) is based on a semi-log form. This represents a problem for those state pairs with zero migration flows, since dropping these observations from the data set could generate selection bias. In the Indian sample, these types of state pairs represent 10% of total observations. One way to avoid sample selection problems arising from excluding observations with zero migration is to add 1 to each bilateral migration rate observation. Nevertheless, the problem remains that the log-linear specification will cause the ordinary least squares (OLS) estimation of elasticities to be inconsistent in the presence of heteroskedasticity in the error term ($u_{ij,t}$).⁷ Santos Silva and Tenreyro (2006) suggest using a PPML estimator with robust standard errors to produce consistent estimates in a nonlinear model. The assumption of equality between the standard deviation and the mean of the dependent variable that is characteristic of the standard Poisson maximum likelihood estimator is no longer necessary in the PPML method. Therefore, we rely on the results from using the PPML estimator.

Finally, multilateral resistance has been identified as a potential source of bias in the application of gravity models (Anderson 2011). It implies that the bilateral migration rate would depend not only on the comparison between the origin and the destination state characteristics but also on the opportunities in all the alternative destinations. The estimating equation is derived based on the assumption that the error terms are distributed according to an extreme value type-1 distribution, which effectively means an assumption of independence from irrelevant alternatives for migration. If this assumption does not hold and there is a need to account for multilateral resistance, Feenstra (2002) suggests that the inclusion of time-varying fixed effects for destination states yields consistent estimates in the presence of multilateral resistance. Bertoli and Fernández-Huertas Moraga (2013) suggest using Pesaran's common correlated effects estimator, which requires a long time span of data. However, this is impossible in our case because our data are based on only two census rounds. Instead, we control for possible multilateral resistance through the inclusion of destination state and time fixed effects.

Given the above discussion, we argue that what we measure in the reduced-form Equation (2) is the total effect of climate variability on bilateral migration, although the underlying mechanism through which climate affects migration is not present in the equation.

4. Data and Measures of Climate Variability

4.1 Definition of migration

A migrant in the Indian Census is defined as an individual with the intent of staying permanently, and a stay in the destination state for at least 6 months; it is a measure of permanent rather than temporary migration. The census identifies migration flows according to the current place of residence (destination state) and the place of residence of provenance (origin state), and includes different durations of stay. We use the 1-year duration to retain a strict separation between the timing of climate variability and migration, and to minimize the measurement error linked to subsequent moves. Our dependent variable is the gross

7 The Breusch–Pagan/Cook–Weisberg test of heteroskedasticity in an OLS regression leads to a test statistic of 368.53 and a p -value of 0. Thus, the null hypothesis of homoskedasticity is rejected.

migration flow $m_{ij,t}$ from state i to state j between year $t-1$ and year t , divided by the population that did not move in the same period, and multiplied by 100,000 to allow for scaling. [Supplementary Appendix Table SA2](#) shows that the average bilateral migration rate is around 8 per 100,000 individuals—which might seem small—but the variable measures the bilateral rate for a unique origin-destination pair in single year, for example 8 of 100,000 individuals migrated from Assam to West Bengal between 1990 and 1991, which is almost 1800 individuals.⁸ We have 930 such combinations. It is important also to note that the dispersion is large (standard deviation almost four times the mean) and that the bilateral migration rate can take values from 0 and to 455 migrants per 100,000 individuals.

4.2 Climate variability: the SPI

Rainfall is the main factor in vulnerability to water availability. Scarcity of water has negative consequences for food availability and human health, and can be the cause of diseases and population displacements (IPCC 2014). In urban areas, the consequences of scarce water supply include difficulty to cover the drinking water requirements in terms of both quantity and quality. In rural areas, output and quality of crops are also affected. The agricultural sector in India is particularly vulnerable to water availability (O'Brien et al. 2004). To test the hypothesis that climate variability acts as a push factor in internal migration, we compute normalized measures of low precipitation and excess precipitation using the Climatic Research Unit (CRU) TS3.21 data set from the University of East Anglia.

The data allow us to calculate the SPI, a frequently used standardized measure of drought developed by McKee et al. (1993). First, a gamma distribution is fitted to the long-run precipitation data (from 1901 to 2001). This then is transformed into a standard normal distribution with zero mean and variance of 1, which gives the SPI. Conceptually, the SPI represents a z -score, or the number of standard deviations of an event above or below the long-run mean. The SPI allows us to determine drought or excess precipitation during a given period in a given location.

The main advantages of the SPI are that it takes account of the spatial and temporal deviations, and measures the start, length, and intensity of a drought or a period of excess precipitation, rather than only the absolute value of precipitation and temperature. It provides a measure with a fixed mean and variance, allowing comparison of the SPIs for different locations. Although the SPI was developed to measure drought, it has been suggested that it is also a good indicator of flood (see for instance Seiler et al. 2002). However, floods can be of different types (e.g. storm surges; flash floods; river floods), and can depend not just on the quantity of rainfall but also on the soil type of flood banks and the topology of the landscape.

The raw data are district-level data; to aggregate them to state level requires calculation of the SPIs of every state. [Supplementary Appendix B](#) provides a principal component analysis to test this procedure. We create three variables based on the SPI to measure the frequency, duration, and magnitude of drought and excess precipitation:

1. *Frequency*: We define a binary variable (by state) that takes the value 1 if there was moderate or severe drought/excess precipitation recorded in a month in that state, and

8 In 1990, 22,408,756 individuals did not move from West Bengal.

0 otherwise.⁹ The frequency measure is the number of months with drought/excess precipitation in the origin state during the 5 years preceding migration, to account for persistence in the effects of drought/excess precipitation.¹⁰ These measures count the total months of either severe or moderate drought/excess precipitation; extreme events are not common in state-level data. Aggregation at state level removes district-level extreme events and can lead to less precise results. More frequent drought/excess precipitation may increase expectations of future similar events, and thus higher frequency should encourage migration.

2. *Maximal duration*: To capture the impact of a long period of drought or excess precipitation, we compute the maximal duration in number of months of such an event during the 5 years preceding migration. Long duration of drought or excess precipitation in a given period is more likely to have a strong negative impact on livelihoods and hence encourage migration to seek better economic conditions.
3. *Magnitude*: This variable is defined as the sum of the absolute values of the SPI for drought or excess precipitation in the 5 years preceding migration. Severe or extreme drought/excess precipitation can affect people by destroying their crops or capital, and having negative effects on health all of which encourage or force migration.

Duration and magnitude are widely used measures of climate variability and two main dimensions of drought or excess precipitation (Zargar et al. 2011). Also, these measures are strictly exogenous and not influenced by economic activity at the time-scale considered here. We constructed and tested additional measures to account for interaction effects such as a long and severe drought; these were never significant and are not included here.

4.3 Other migration determinants

Since climatic factors are not the only determinants of migration, we control also for the most important social and economic drivers by estimating bilateral migration rates as a function of distance, common border, common language, bilateral migrant networks, the rate of SCs and STs in the origin state, and a set of fixed effects. Climate-dependent explanatory variables, such as income and agricultural income per capita, irrigation, and urbanization rates, are discussed in Section 5.1.3, where we explore channels of the indirect effect of climate variability on inter-state migration. [Supplementary Appendix A](#) describes the measures, data sources, and descriptive statistics.

5. Results

We start by presenting the OLS estimates of [Equation \(2\)](#) with bilateral fixed effects instead of distance, common border, and common language, to capture all potential time-invariant factors that may affect bilateral migration. The estimation results in [Table 1](#) indicate that

- 9 See definition of moderate and severe drought/excess precipitation in [Supplementary Appendix Table SA1](#).
- 10 In an analysis of the impact of drought on rural wages in Brazil, [Mueller and Osgood \(2009\)](#) identify a 5-year persistence effect from drought. [Barrios et al. \(2006\)](#) and [Strobl and Valfort \(2013\)](#) also use a lag of 5 years for the impact of natural disasters and climate variables. Estimations in [Supplementary Appendix Table SD6](#) show the results with different lags.

Table 1. Inter-state migration and drought with bilateral fixed effects in an OLS model

	(1)	(2)	(3)
$\ln SC_{it}$	-7.220 (12.675)	-0.530 (12.612)	-5.197 (12.657)
$\ln ST_{it}$	-2.023 (8.778)	-0.515 (8.865)	-1.689 (8.847)
$\text{Network_rate}_{ijt}$	0.198 (0.175)	0.222 (0.176)	0.224 (0.173)
Drought frequency _{it}	0.017** (0.007)		
Longest drought dur _{it}		0.013* (0.007)	
Drought magnitude _{it}			0.008 (0.005)
Origin-state FE	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes
Bilateral FE	Yes	Yes	Yes
N	1860	1860	1860
R ²	0.828	0.828	0.827

Note: The dependent variable is $\ln(\text{bilateral migration rate} + 1)$ from state i to state j between year $t-1$ and year t . The subscript t indicates only that the variable varies over time. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

one additional month of drought increases the bilateral migration rate by 1.7% at the 5% level of significance (Column (1)). The longest drought duration (Column (2)) is weakly significant (10% level). Drought magnitude (Column (3)) and the other covariates are not significant, and the coefficient estimates may be inconsistent in the presence of heteroskedasticity (Section 3.2). To illustrate the effect of using the PPML estimator, we compare the results using OLS and PPML (Supplementary Appendix Table SD1). In general, the results vary much between the two estimators. In the full sample (Columns (1) and (3)), the effect of drought frequency goes from 1.7% with OLS to 1.5% with the PPML estimator and with a decrease in significance. Since PPML gives consistent estimates, we present all the following results with the PPML estimator, accounting for zero observations, unless otherwise stated. All the estimations include origin-state fixed effects and destination-time fixed effects, but not bilateral fixed effects, since adding too many explanatory variables creates convergence problems when using the PPML estimator.

Climate variability can have both a direct (amenity effect) and an indirect effect on internal migration. Section 5.1 analyzes the total effect of climate variability in terms of drought, and Section 5.1.2 presents the different migration responses. Section 5.1.3 discusses the potential channels underlying the results. Section 5.2 presents the results from the measures of excess precipitation, and Section 5.3 presents some robustness tests using alternative measures of climate variability and a different econometric specification. In Section 5.4 we calculate the magnitude of the migration flows induced by drought variability over the period studied.

5.1 Drought and migration

Table 2 presents the main estimation results of Equation (2) for the drought measures. The different measures are introduced separately in the estimations because of their correlation (see Supplementary Appendix Table SA3).

The results show that the proxies for the costs of migration are the most important factors in internal migration in terms of value and statistical significance. Bilateral migration rates between contiguous states are 2.4 times higher than for states with no common border. States with a common language have 50% higher bilateral migration rates.¹¹ Geographical distance is also statistically significant with a 1% larger distance decreasing the bilateral migration rate by 0.7%. A 1 percentage point increase in the migrant network rate increases the bilateral migration rate by 6.4%, which is in line with the findings in the literature on migrant networks (Beine et al. 2011). The SC and ST rates in the origin state are not significant.

Among the three drought measures tested, the duration of the longest drought is rejected as a push factor for migration. The results indicate that an additional month of drought during the 5 years preceding migration increases the bilateral migration rate by 1.5% (Column (1)) and that an additional 1-unit increase (which is very high) in absolute magnitude in the SPI increases the migration rate by 0.8% (Column (3)). The statistical significance of drought frequency is higher than that of the magnitude of droughts: 5.1 and 10.0%, respectively.

5.1.1 The timing of climatic factors and migration

One of the advantages of our study is the fact that our data allow us to measure climate variability before the migration decision. This contrasts with other work, especially on international migration, where data constrain the analysis to use average climate variability and average migration over the same 5- or 10-year periods. To compare our results with the method used in the literature, we use similar measures of climate variability, that is temperature and rainfall deviations in absolute value from their long-run mean, and anomalies defined as in Marchiori et al. (2012).¹² We also separate positive and negative anomalies to measure excess or deficit temperature and precipitation.

Table 3 presents the difference in results using contemporaneous climate variability compared to climate variability averaged over a longer time span. The dependent variable in Columns (1)–(4) is the average bilateral migration rate between 1982 and 1991 for the 1991 census, and between 1992 and 2001 for the 2001 census. The climate variability measures are defined as temperature and precipitation anomalies and deviations averaged over the same 10-year periods. The dependent variable in Columns (5)–(8) is our 1-year bilateral migration rate, with the climate variability measured in the 5 years preceding migration.

Columns (1) and (5) present the results with positive temperature and negative precipitation anomalies. Columns (2) and (6) present the results with negative temperature and positive precipitation anomalies. Columns (3) and (7), and Columns (4) and (8) present the results with the corresponding measures of temperature and precipitation deviations,

11 The marginal effects of the dummy variables are calculated as $(e^{b_i} - 1)$, where b_i is the estimated coefficient of the variable.

12 The definition is provided in Supplementary Appendix A.

Table 2. Inter-state migration and drought: total effect

	(1)	(2)	(3)
$\ln \text{ distance}_{ij}$	−0.658*** (0.080)	−0.658*** (0.080)	−0.658*** (0.080)
Border_{ij}	1.226*** (0.150)	1.221*** (0.150)	1.222*** (0.149)
Language_{ij}	0.377** (0.160)	0.376** (0.161)	0.376** (0.162)
$\ln SC_{it}$	−9.682 (18.386)	−1.710 (18.476)	−4.360 (18.472)
$\ln ST_{it}$	1.540 (6.352)	2.616 (6.261)	1.756 (6.288)
$\text{Network_rate}_{ijt}$	0.064** (0.020)	0.064** (0.020)	0.064** (0.020)
$\text{Drought frequency}_{it}$	0.015* (0.008)		
$\text{Longest drought dur}_{it}$		0.010 (0.007)	
$\text{Drought magnitude}_{it}$			0.008* (0.005)
Origin-state FE	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes
N	1860	1860	1860
R^2	0.698	0.695	0.694

Note: The dependent variable is the bilateral migration rate from state i to state j between year $t-1$ and year t . The subscript t indicates only that the variable varies over time. Robust standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

respectively. The results show that for the 10-year averaged migration measure, only positive deviations in precipitation have a weakly significant effect on migration rates, and this effect is negative. Comparing this to the measures of climate variability during the 5 years before the 1-year migration flow, we find that negative precipitation anomalies have a positive impact on the bilateral migration rate (Column (5)) similar to our drought measure, and that positive precipitation anomalies (Column (6)) decrease the bilateral migration rate. This indicates that excess precipitation in this sense is favorable and not equivalent to flood. Qualitatively similar results are obtained using the deviations measures. These results support the improvement in the estimations from measuring the effect with appropriate timing of the climate variability measures and migration resulting from use of census data with a 1-year duration of migration.

5.1.2 Heterogeneous effects

In this section, we show the different migratory responses to drought depending on the level of agricultural activity in the state, irrigation, gender, and origin and destination of migration flows.

Table 3. Inter-state migration and climate anomalies: 10-year average comparison

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Migration rate	10-year average			Migration rate	1-year flow		
$\ln distance_{ij}$	-0.716*** (0.074)	-0.714*** (0.074)	-0.716*** (0.074)	-0.716*** (0.074)	-0.657*** (0.079)	-0.658*** (0.080)	-0.657*** (0.080)	-0.657*** (0.080)
$Border_{ij}$	1.424*** (0.148)	1.420*** (0.148)	1.422*** (0.149)	1.420*** (0.148)	1.222*** (0.149)	1.223*** (0.148)	1.222*** (0.149)	1.223*** (0.148)
$Language_{ij}$	-0.013 (0.147)	-0.009 (0.146)	-0.011 (0.146)	-0.010 (0.147)	0.384** (0.161)	0.390** (0.159)	0.384** (0.161)	0.387** (0.160)
$\ln SC_{it}$	-2.050 (15.672)	-8.376 (16.172)	-0.990 (15.759)	-4.761 (15.658)	-3.382 (18.444)	-16.151 (18.713)	-3.411 (18.347)	-8.526 (18.247)
$\ln ST_{it}$	9.887 (6.730)	10.658 (6.681)	9.902 (6.778)	10.298 (6.759)	1.540 (6.217)	2.304 (6.168)	1.820 (6.256)	2.114 (6.242)
$network_rate_{it}$	0.067*** (0.020)	0.065** (0.021)	0.067*** (0.020)	0.064** (0.021)	0.064** (0.020)	0.063** (0.021)	0.064** (0.020)	0.062** (0.021)
Precipitation (-) anomaly _{it}	1.611 (1.131)				2.545** (1.022)			
Temperature (+) anomaly _{it}	-3.264 (2.446)				-2.227 (2.351)			
Precipitation (+) anomaly _{it}		-0.621 (0.420)				-1.104** (0.392)		

(continued)

Table 3. (continued)

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Migration rate	10-year average			Migration rate	1-year flow		
Temperature (−) anomaly _{it}		0.136 (0.991)				−0.657 (0.882)		
Precipitation (−) deviation _{it}			0.005 (0.003)				0.006** (0.003)	
Temperature (+) deviation _{it}			−0.475 (0.763)				−0.136 (0.738)	
Precipitation (+) deviation _{it}				−0.002* (0.001)				−0.003** (0.001)
Temperature (−) deviation _{it}				−0.050 (0.624)				−0.076 (0.490)
Origin-state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1860	1860	1860	1860	1860	1860	1860	1860
R ²	0.761	0.759	0.760	0.760	0.696	0.698	0.694	0.697

Note: The dependent variable is the bilateral migration rate from state *i* to state *j*. The subscript *t* indicates only that the variable varies over time. Robust standard errors in parentheses. **p*<0.10, ***p*<0.05, ****p*<0.01.

Table 4. Inter-state migration and drought in agricultural states

	(1)	(2)	(3)
$\ln \text{ distance}_{ij}$	-0.729*** (0.077)	-0.730*** (0.076)	-0.729*** (0.076)
Border_{ij}	1.099*** (0.116)	1.099*** (0.116)	1.101*** (0.116)
Language_{ij}	0.037 (0.130)	0.033 (0.131)	0.036 (0.130)
$\ln SC_{it}$	3.103 (14.632)	2.188 (15.082)	3.898 (14.490)
$\ln ST_{it}$	10.785 (10.529)	9.450 (10.777)	9.055 (10.496)
$\text{Network_rate}_{ijt}$	0.081*** (0.019)	0.081*** (0.019)	0.081*** (0.019)
$\text{drought freq}_{it}(\text{AgrSt})$	0.017** (0.007)		
$\text{longest drought dur}_{it}(\text{AgrSt})$		0.003 (0.006)	
$\text{drought magnitude}_{it}(\text{AgrSt})$			0.012** (0.005)
Origin-state FE	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes
N	1560	1560	1560
R^2	0.672	0.668	0.672

Note: The dependent variable is the bilateral migration rate from state i to state j between year $t-1$ and year t . The subscript t indicates only that the variable varies over time. Robust standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.1.2.1 Extent of agriculture in the state economy. To test for a heterogeneous effect in agricultural states, we introduce an interaction term with an agricultural dummy variable that takes the value 1 if the agricultural Net State Domestic Product (NSDP) exceeds the median value among the states (Table 4 Columns (1)–(3)).¹³ The sample size is smaller for these estimations, since agricultural NSDP data are not available for four union territories and one state (Chandigarh, Dadra and Nagar Haveli, Daman and Diu, Lakshadweep, and Mizoram). In these estimations, common language no longer affects the bilateral migration rate, and the effect of network is higher. The effect of drought frequency is stronger and more significant in agricultural states, with each additional month of drought inducing an increase in the bilateral migration rate of 1.7% if the origin state is agricultural. In agricultural states of origin, 1 additional unit increase in the magnitude of drought (measured by the SPI in absolute magnitude) implies a 1.2% increase in the average bilateral migration rate. In agricultural states, the two effects are significant at the 5% level.

13 The 13 states with an agricultural NSDP per capita higher than the median are Arunachal Pradesh, Assam, Bihar, Haryana, Himachal Pradesh, Madhya Pradesh, Orissa, Punjab, Rajasthan, Sikkim, Tripura, Uttar Pradesh, and Andaman and Nicobar Islands.

5.1.2.2 Irrigation as an adaptation measure. Other adaptations than migration can limit the impact of climate variability (Barnett and Webber 2010; Mendelsohn 2012). In agriculture, farmers can adapt to shortfalls in precipitation or to increased variability in precipitation by changing to more resistant crops, or by investing in irrigation infrastructure (O'Brien et al. 2004). Here, the analysis is at the macro level, and we cannot control for drought-resistant crops. Although irrigation is likely to be dependent on climatic factors on its own, we control for irrigation capacity as one of the most common adaptation measures against drought. Taraz (2015) finds that Indian farmers adjust their irrigation investment according to monsoon rainfall variability, but that its efficacy in reducing the losses in agricultural profits is limited. To test the effect of irrigation, we use the ratio of net irrigated land in total cultivated land in the origin state. [Supplementary Appendix Table SD2](#) shows the net effect of drought frequency including the net irrigation rate and the interaction terms. The interaction terms between the drought measures and irrigation have the expected negative sign but are never significant (Columns (4)–(6)). Since irrigation is correlated to climate (see [Supplementary Appendix Table SA3](#)) and not on its own a determinant of migration, because of multicollinearity, its inclusion will only reduce the precision of the estimated coefficient of drought. We observe that the significance and magnitude of the coefficients of drought are attenuated if the net irrigation measure is included (Columns (1)–(3)). Nevertheless, the effect of drought frequency maintains its sign and order of magnitude, which confirms the robustness of its effect.

5.1.2.3 Male and female migration rates. The Indian Census asks individuals to indicate the reason for migration from a list of work/employment, business, education, marriage, moved after birth, moved with household, and others. [Supplementary Appendix Table SC2](#) shows that the family moving is the main reason for migration among women (41% of women in 1991 and 48% of women in 2001), and employment is the main reason for men (42% of men in 1991 and 54% of men in 2001). To further test the relationship between climate variability and internal migration in India, we run separate estimations on male and female migration rates. [Table 5](#) reports the results for male migration (Columns (1)–(3)) and female migration (Columns (4)–(6)). In the case of male migration, all the determinants have similar size and significance as in the main estimations of total migration rates in [Table 2](#). Most importantly, drought frequency, duration, and magnitude have the same marginal effect, but drought frequency has a higher level of significance. In the case of female migration, the results are similar to the main estimations of total inter-state migration but show a lower significance.

5.1.2.4 Different flows according to source and destination. So far, we have studied total inter-state migration flows. However, the analysis of the precipitation data in [Supplementary Appendix B](#) shows that aggregation at state level masks important variability among districts. Detailed modeling of rural–urban migration flows at district level would add to our understanding of the relation between climate variability and migration; however, the census data do not allow this, since origin districts are not recorded. Nevertheless, it is possible within inter-state migration flows to distinguish whether the origin and destination are rural or urban. Note that in India, rural–rural migration is more frequent than rural–urban migration, as shown in [Supplementary Appendix Table SC3](#). We use the information on inter-state migration flows between areas characterized as rural or

Table 5. Inter-state male and female migration and drought

	(1)	(2)	(3)	(4)	(5)	(6)
	Male	Male	Male	Female	Female	Female
$\ln \text{ distance}_{ij}$	−0.683*** (0.084)	−0.683*** (0.083)	−0.683*** (0.083)	−0.624*** (0.081)	−0.624*** (0.080)	−0.624*** (0.081)
Border_{ij}	1.099*** (0.152)	1.094*** (0.152)	1.095*** (0.152)	1.394*** (0.151)	1.390*** (0.150)	1.391*** (0.151)
Language_{ij}	0.465** (0.164)	0.465** (0.165)	0.464** (0.166)	0.257 (0.161)	0.254 (0.162)	0.255 (0.162)
$\ln \text{ SC}_{it}$	−11.628 (19.364)	−3.782 (19.607)	−6.731 (19.574)	−6.068 (17.341)	1.947 (17.071)	−0.271 (17.102)
$\ln \text{ ST}_{it}$	1.091 (6.459)	2.145 (6.412)	1.254 (6.437)	2.500 (6.609)	3.584 (6.476)	2.780 (6.503)
$\text{Network_rate}_{ijt}$	0.061** (0.021)	0.061** (0.021)	0.061** (0.021)	0.068*** (0.020)	0.068*** (0.020)	0.068*** (0.020)
$\text{Drought frequency}_{it}$	0.015** (0.007)			0.015* (0.009)		
$\text{Longest drought dur}_{it}$		0.010 (0.006)			0.009 (0.007)	
$\text{Drought magnitude}_{it}$			0.008* (0.005)			0.007 (0.005)
Origin-state FE	Yes	Yes	Yes	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1860	1860	1860	1860	1860	1860
R^2	0.673	0.669	0.668	0.723	0.722	0.721

Note: The dependent variable is the bilateral migration rate from state i to state j between year $t-1$ and state t , separately for male and female migrants. The subscript t indicates only that the variable varies over time. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

urban to analyze the patterns of migration in more detail. The results are reported in Table 6. The dependent variable is the bilateral migration rate from one part of a state to one part of another state (e.g. from a rural to an urban area); we therefore exclude the independent variables for the entire state (e.g. SC or ST rates in the origin state, and bilateral migrant networks). We test separately for an effect on total rural out-migration to another state (rural and urban destinations together), rural–urban migration, and rural bilateral migration (rural origin and rural destination involving different states).¹⁴

Analysis of these three patterns of bilateral migration confirms the hypothesis of drought frequency as a push factor in migration. The results in Table 6 Column (1) show that drought frequency has a higher impact on total rural out-migration than on total inter-state migration: its average effect increases from 1.5 to 2%. Table 6 Column (2) shows that drought frequency has a positive and significant impact on rural–urban migration, but that

14 We show the results on drought frequency, since it has the strongest effect on bilateral migration. The other measures—drought duration and magnitude—are not significant at this disaggregated level.

Table 6. Rural out-migration and drought frequency

Migration pattern	(1) Rural–total	(2) Rural–urban	(3) Rural–rural
$\ln \text{ distance}_{ij}$	−0.702*** (0.096)	−0.663*** (0.089)	−0.740*** (0.114)
Border_{ij}	1.263*** (0.175)	1.201*** (0.187)	1.341*** (0.202)
Language_{ij}	0.444** (0.190)	0.712*** (0.192)	0.215 (0.201)
$\text{Drought frequency}_{it}$	0.020** (0.010)	0.017* (0.009)	0.025** (0.011)
Origin-state FE	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes
N	1860	1860	1860
R^2	0.815	0.813	0.751

Note: The dependent variable is the bilateral migration rate from zones in state i to state j between year $t-1$ and year t . The subscript t indicates only that the variable varies over time. Robust standard errors in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

the effect is smaller than the effect on total rural out-migration. Column (3) shows that the effect of drought frequency is the largest on the rural–rural bilateral migration flows at 2.5%. Analysis of the separate inter-state migration flows according to rural out-migration supports the robustness of our main estimations using total inter-state migration flows presented in Table 2, and suggests that an important part of the impact of climate variability on migration is its impact on rural areas.

5.1.3 Climate variability and migration: What are the channels?

We have estimated the total (both direct and indirect) effect of drought on internal migration in India. The indirect effects could work through the effects of climate variability on average income (Beine and Parsons 2015), agricultural income or yield (Feng et al. 2012; Viswanathan and Kumar 2015), urbanization (Barrios et al. 2006), and conflict (Wischnath and Buhaug 2014).¹⁵ This section explores the contribution made by three potential channels to explaining the effect of drought frequency on bilateral inter-state migration rates.

5.1.3.1 *Income channel.* Table 7 presents the same estimations as in Table 2 Columns (1)–(3), but in Columns (2)–(4) adds the ratio of NSDP per capita between the destination and origin states. NSDP per capita in the destination state is a proxy for the average income expected by migrants. Column (1) presents the estimation with the income per capita ratio and without the variables for climate variability. In all the estimations, the income ratio is positive and significant, implying that migrants migrate to states where the expected income is higher than in the origin state. The income ratio elasticity ranges from 0.6 to 0.9. If

15 We performed estimations for the conflict channel using data on homicide rates from the Indian Ministry of Home Affairs, National Crime Records Bureau, but found no effect.

Table 7. Inter-state migration and drought: total income and agricultural channel

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln \frac{y_{it}}{y_{jt}}$	0.877** (0.398)	0.675* (0.371)	0.795** (0.391)	0.756* (0.393)				
$\ln \frac{y_{agr,it}}{y_{agr,jt}}$					0.178 (0.233)	0.203 (0.233)	0.181 (0.235)	0.201 (0.237)
$\ln \text{distance}_{ij}$	-0.658*** (0.080)	-0.658*** (0.081)	-0.658*** (0.080)	-0.658*** (0.080)	-0.693*** (0.080)	-0.693*** (0.080)	-0.693*** (0.080)	-0.692*** (0.080)
Border_{ij}	1.224*** (0.149)	1.228*** (0.149)	1.225*** (0.149)	1.225*** (0.149)	1.127*** (0.124)	1.129*** (0.124)	1.127*** (0.124)	1.129*** (0.124)
Language_{ij}	0.379** (0.161)	0.379** (0.159)	0.378** (0.160)	0.378** (0.160)	0.120 (0.136)	0.121 (0.133)	0.120 (0.135)	0.121 (0.133)
$\ln SC_{it}$	-0.454 (18.318)	-5.010 (18.423)	2.189 (18.591)	-0.267 (18.461)	7.271 (15.111)	4.182 (14.848)	8.005 (15.509)	5.102 (14.812)
$\ln ST_{it}$	6.998 (6.685)	5.439 (6.822)	6.986 (6.727)	6.125 (6.783)	10.297 (10.549)	15.682 (10.225)	11.344 (10.589)	11.902 (10.565)
network_rate_{it}	0.064** (0.021)	0.064** (0.020)	0.063** (0.020)	0.063** (0.020)	0.082** (0.026)	0.082** (0.025)	0.082** (0.026)	0.081** (0.026)
Drought frequency _{it}		0.013* (0.007)				0.008* (0.005)		
Longest drought dur _{it}			0.008 (0.007)				0.002 (0.005)	
Drought magnitude _{it}				0.005 (0.005)				0.005 (0.004)
Origin-state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1860	1860	1860	1860	1300	1300	1300	1300
R ²	0.694	0.699	0.697	0.696	0.672	0.677	0.673	0.677

Note: The dependent variable is the bilateral migration rate from state i to state j between year $t-1$ and year t . The subscript t indicates only that the variable varies over time. Robust standard errors in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the income channel captures all the effects of climate variability, the coefficients of the climate variability should be nonsignificant in Columns (2)–(4). If we compare Columns (1) and (2), we observe that the significance and magnitude of both average income ratio and drought frequency decrease in Column (2). This change in significance is observed also for drought magnitude (Column (4)). These results suggest that part of the effect of climate variability on migration goes through income. In particular, drought magnitude is captured entirely by the indirect effect on income, but in the case of drought frequency, there is also a direct effect which is smaller—1.3%—and significant only at the 9.3% level.

These effects are explained by the direct impact of drought frequency on income. Columns (1)–(3) in [Supplementary Appendix Table SD3](#) present the estimations of the income ratio on drought. The three drought variables (frequency, duration, and magnitude) have a positive and statistically significant impact on the income ratio. More frequent, longer and severe droughts in the origin state increase the difference in incomes between the destination and the origin states and thus might encourage migration indirectly. Given that all the variables for drought in the origin state have a highly significant impact on the income ratio, we should exclude income ratio from the estimations of bilateral migration rates as in [Table 2](#), to avoid bad control.

5.1.3.2 Agricultural channel. The effects of climate change on agricultural yields in India have been documented extensively ([Guiteras 2009](#); [Krishnamurthy 2012](#)). [Table 7](#) Columns (5)–(8) present the results for the agricultural channel. The analysis is similar to the analysis of average income per capita; we include the ratio of per capita agricultural income in the destination state to per capita agricultural income in the origin state.¹⁶ The results show that the *agricultural* income ratio is not significant for determining bilateral migration rates (Column (5)) which depend on *total* per capita income in the destination versus the origin states. Both pull factors of nonagricultural employment and push factors of decreasing agricultural income are at work. When controlling for the ratio of agricultural income between the destination and origin states, the effect of another month of drought decreases to 0.8% (significant at the 8.5% level only) indicating that the effect of climate variability in the origin state is largely mediated by agricultural income. [Supplementary Appendix Table SD3](#) Columns (4)–(6) show that agricultural income in the origin state is significantly and negatively affected by all three drought measures. The results in [Tables 7](#) and [Supplementary Appendix Table SD3](#) confirm that part of the effect of climate variability on migration in India goes through the agricultural channel.

To analyze this channel further, we compare our results with the ones obtained in [Viswanathan and Kumar \(2015\)](#) on inter-state out-migration and agricultural income. They find an elasticity of out-migration with respect to agricultural income of -0.775 . The comparison is not straightforward, since [Viswanathan and Kumar \(2015\)](#) (i) analyzes 15 major states;¹⁷ (ii) uses out-migration rates only at state level; (iii) controls for temperature

16 The sample size is smaller for these estimations, since data on agricultural NSDP are not available for four union territories and one state. Using a ratio decreases the sample size further, from 1560 to 1300, because of missing values in either the origin or destination states which might explain why the agricultural income ratio is not significant in these estimations.

17 The 15 major states analyzed by [Viswanathan and Kumar \(2015\)](#) are Andhra Pradesh, Bihar, Gujarat, Haryana, Himachal Pradesh, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Orissa, Punjab, Rajasthan, Tamil Nadu, Uttar Pradesh, and West Bengal.

and rainfall in absolute levels, and include no other control variables except fixed effects in the origin state; and (iv) takes account only of rural out-migration for male migrants declaring work as the reason for migration. For our sample of states for which we have agricultural NSDP data ($n = 1560$), and using only rural out-migration flows as in Viswanathan and Kumar (2015), the estimations presented in [Supplementary Appendix Table SD4](#) give an elasticity of -0.105 without controlling for fixed effects (Column (3)), and an elasticity of -0.075 that is not significant when controlling for origin-state and destination-time fixed effects and for temperature and precipitation anomalies (Column (4)). We ran other estimations that were as similar as possible to the state-level estimations in Viswanathan and Kumar (2015) to allow for a comparison with the results for elasticity of migration with respect to agricultural income. [Supplementary Appendix Table SD4](#) Columns (1) and (2) present the estimations of the bilateral migration rates using the migration between the same 15 states ($n = 420$); the estimated elasticity is -0.2 with fixed effects, and is not significant. The estimated elasticities with respect to agricultural income on bilateral migration rates we obtained are smaller than the estimated elasticity in Viswanathan and Kumar (2015).

5.1.3.3 Urbanization channel. It is possible that the impact of climate variability on migration is due mainly to rural–urban migration, and thus urbanization. To study this channel, we control for the urbanization rate in the state of origin to proxy for alternative opportunities to inter-state migration. Following drought, a more urbanized origin state should offer more alternative nonagricultural employment probabilities than a less urbanized origin state. [Table 8](#) shows no significant effect of the urbanization rate in the state of origin on the bilateral migration rate. Moreover, the effects of drought frequency and magnitude remain the same as in [Table 2](#). Thus, we find no evidence of an urbanization channel affecting inter-state migration in India.

This does not mean that climate variability has no effect on the urbanization rate in the origin state. [Supplementary Appendix Table SD5](#) Columns (1)–(3) show a positive and significant effect of drought frequency, duration, and magnitude on the urbanization rate. These results are in line with the findings in Barrios et al. (2006) for Sub-Saharan Africa. For example, an additional month of drought is associated with an increase of 0.2% in the urban population rate. However, the potential channel of the effect of climate variability passing through urbanization can be rejected, since a direct effect of drought frequency and magnitude remains in the estimations of bilateral migration rates, even when controlling for urbanization.

The results discussed above show that drought has a significant effect on urbanization in the origin state, but that the effects of drought on migration work through income, and mainly through the agricultural sector.

5.2 Excess precipitation and migration

[Table 9](#) presents the estimations in [Tables 2](#) and [4](#) using measures for excess precipitation instead of drought. The marginal effects of the frequency, duration, and magnitude of excess precipitation are negative with a level of statistical significance for frequency and magnitude of 5.5 and 7.3%, respectively. In agricultural states, the impact of the duration of an episode of excess precipitation is strongly significant, and reduces bilateral migration rates

Table 8. Inter-state migration and drought: urbanization channel

	(1)	(2)	(3)	(4)
ln urban_rate _{it}	−0.263 (0.535)	−0.386 (0.523)	−0.258 (0.529)	−0.413 (0.536)
ln distance _{ij}	−0.657*** (0.080)	−0.657*** (0.081)	−0.657*** (0.080)	−0.657*** (0.081)
Border _{ij}	1.219*** (0.149)	1.227*** (0.150)	1.221*** (0.150)	1.223*** (0.150)
Language _{ij}	0.377** (0.163)	0.378** (0.160)	0.376** (0.162)	0.376** (0.162)
ln SC _{it}	−5.088 (18.343)	−8.924 (18.486)	−1.110 (18.580)	−3.211 (18.653)
ln ST _{it}	−0.026 (6.737)	−1.499 (7.016)	0.598 (6.822)	−1.512 (6.887)
Network_rate _{ijt}	0.064** (0.020)	0.064** (0.020)	0.064** (0.020)	0.064** (0.020)
Drought frequency _{it}		0.015** (0.008)		
Longest drought dur _{it}			0.010 (0.007)	
Drought magnitude _{it}				0.009* (0.005)
Origin-state FE	Yes	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes	Yes
N	1860	1860	1860	1860
R ²	0.691	0.698	0.695	0.694

Note: The dependent variable is the bilateral migration rate from state *i* to state *j* between year *t* − 1 and year *t*. The subscript *t* indicates only that the variable varies over time. Robust standard errors in parentheses. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

from these states by 1.5%.¹⁸ The negative impact of excess precipitation can be explained by several factors. First, the measures we use are based on the SPI which is a reliable indicator of drought but a less direct measure of flood, since it captures only climatological floods and not other factors, such as topology and hydrology. Guiteras et al. (2015) compare precipitation data with remote-sensing data on actual flooding in Bangladesh and argue that precipitation data are a weak proxy for floods. We address this concern in Section 5.3.1, where we test alternative flood measures. Second, evidence from other countries, notably Bangladesh (Gray and Mueller, 2012), shows that floods do not always induce migration. Drought can be characterized as a long-run process that does not always induce an immediate response, but when it does may lead to permanent migration. In contrast, flooding is a rapid onset phenomenon which may lead only to short-distance displacement (Barnett and

18 The frequency of excess precipitation retains its negative sign also for rural out-migration. As in the case of drought, the effect is stronger on rural–rural migration but is nonsignificant for rural–urban migration (estimations not shown here, but available on request).

Table 9. Inter-state migration and excess precipitation: total effect and agricultural state effect

	(1)	(2)	(3)	(4)	(5)	(6)
$\ln \text{ distance}_{ij}$	−0.660*** (0.079)	−0.658*** (0.079)	−0.661*** (0.079)	−0.730*** (0.076)	−0.731*** (0.076)	−0.730*** (0.076)
Border_{ij}	1.222*** (0.149)	1.219*** (0.149)	1.223*** (0.149)	1.099*** (0.117)	1.097*** (0.116)	1.099*** (0.116)
Language_{ij}	0.376** (0.160)	0.376** (0.163)	0.375** (0.161)	0.032 (0.131)	0.035 (0.129)	0.032 (0.132)
$\ln SC_{it}$	−20.868 (19.835)	−5.484 (18.329)	−20.927 (20.533)	−0.662 (14.474)	5.103 (14.369)	0.440 (14.519)
$\ln ST_{it}$	3.316 (6.222)	2.286 (6.317)	2.732 (6.229)	6.305 (10.898)	7.604 (10.444)	8.903 (10.872)
$\text{Network_rate}_{ijt}$	0.064** (0.020)	0.064** (0.020)	0.065** (0.020)	0.082*** (0.019)	0.079*** (0.020)	0.081*** (0.020)
$\text{Flood frequency}_{it}$	−0.014* (0.007)					
$\text{Longest flood dur}_{it}$		−0.002 (0.006)				
$\text{Flood magnitude}_{it}$			−0.010* (0.005)			
$\text{flood frequency}_{it}(\text{AgrSt})$				−0.015* (0.009)		
$\text{longest flood dur}_{it}(\text{AgrSt})$					−0.015** (0.007)	
$\text{flood magnitude}_{it}(\text{AgrSt})$						0.000 (0.005)
Origin-state FE	Yes	Yes	Yes	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1860	1860	1860	1560	1560	1560
R ²	0.699	0.691	0.698	0.668	0.675	0.668

Note: The dependent variable is the bilateral migration rate from state i to state j between year $t-1$ and year t . The subscript t indicates only that the variable varies over time. Robust standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Webber 2010; Piguet 2010). Thus, responses to flood events are different, and if migration occurs, it may be temporary (Perch-Nielsen et al. 2008).

5.3 Robustness tests

5.3.1 Additional measures of climate variability

We test several alternative measures of climate variability to improve the robustness of our results. First, we test the long-run temperature and precipitation anomaly measures used in Table 3. Table 10 Column (1) shows the same estimations as in Table 2 with drought frequency. Column (2) includes positive temperature anomaly to control for temperature in addition to negative precipitation anomalies as measured by drought frequency. All the

Table 10. Inter-state migration and long run anomalies in temperature and precipitation

	(1)	(2)	(3)	(4)
$\ln \text{ distance}_{ij}$	−0.658*** (0.080)	−0.658*** (0.081)	−0.658*** (0.079)	−0.657*** (0.080)
Border_{ij}	1.226*** (0.150)	1.226*** (0.150)	1.220*** (0.149)	1.220*** (0.149)
Language_{ij}	0.377** (0.160)	0.377** (0.160)	0.375** (0.163)	0.385** (0.161)
$\ln SC_{it}$	−9.682 (18.386)	−9.725 (18.357)	−5.972 (18.284)	−3.153 (18.393)
$\ln ST_{it}$	1.540 (6.352)	1.549 (6.358)	1.956 (6.245)	1.667 (6.231)
$\text{Network_rate}_{ijt}$	0.064** (0.020)	0.064** (0.020)	0.064** (0.020)	0.064** (0.020)
$\text{Drought frequency}_{it}$	0.015* (0.008)	0.015* (0.008)		
$\text{Temperature (+) anomaly}_{it}$		0.552 (2.466)	−1.565 (2.317)	
$\text{Precipitation (−) anomaly}_{it}$				2.407** (0.986)
Origin-state FE	Yes	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes	Yes
N	1860	1860	1860	1860
R^2	0.698	0.698	0.692	0.695

Note: The dependent variable is the bilateral migration rate from state i to state j between year $t-1$ and year t . The subscript t indicates only that the variable varies over time. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

effects—including drought frequency—remain stable, but the temperature anomaly is not statistically significant. Column (3) includes only the positive temperature anomaly. Its effect is negative and not significant. In Column (4) the negative precipitation anomaly replaces the temperature variable. The coefficient is positive, indicating that a deficit in precipitation will increase migration which is in line with the previously reported results on drought measures based on the SPI. Two interesting conclusions can be drawn from these results. First, there is no evidence of omitted variables bias from including only precipitation-based measures and not temperature (Auffhammer et al. 2013). Second, unlike the precipitation variables, the temperature variables are not significant or stable in either magnitude or direction.

Table 11 tests additional climate variability measures. Columns (1)–(3) present the continuous values of the SPI directly. In Column (1), the measure is the value of the annual SPI average 5 years before migration which is aimed at capturing average deviations (positive or negative) in climate variability without distinguishing between positive and negative shocks. The large size of most Indian states limits this measure, however, and we find no statistical significance. Columns (2) and (3) present SPIs superior to +1 and inferior to −1 (in absolute values) to capture positive (excess precipitation) or negative (drought) shocks

Table 11. Inter-state migration and other alternative climate variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln \text{ distance}_{ij}$	-0.658*** (0.079)	-0.658*** (0.079)	-0.658*** (0.080)	-0.657*** (0.079)	-0.658*** (0.079)	-0.658*** (0.079)	-0.658*** (0.080)	-0.658*** (0.080)
Border_{ij}	1.219*** (0.149)	1.219*** (0.149)	1.221*** (0.149)	1.218*** (0.149)	1.218*** (0.149)	1.217*** (0.149)	1.220*** (0.149)	1.222*** (0.149)
Language_{ij}	0.375*** (0.163)	0.376*** (0.163)	0.375*** (0.161)	0.378*** (0.162)	0.377*** (0.162)	0.376*** (0.162)	0.379*** (0.162)	0.384*** (0.160)
$\ln SC_{it}$	-5.361 (17.957)	-6.021 (18.807)	-1.685 (18.346)	-5.156 (18.267)	-5.640 (18.200)	-5.713 (18.195)	-4.135 (18.406)	-2.650 (18.512)
$\ln ST_{it}$	2.063 (6.237)	2.101 (6.215)	3.002 (6.261)	1.596 (6.250)	1.531 (6.266)	1.282 (6.283)	2.414 (6.259)	1.869 (6.294)
Network_rate_{it}	0.064*** (0.020)	0.064*** (0.020)	0.064*** (0.020)	0.065*** (0.020)	0.064*** (0.020)	0.064*** (0.020)	0.064*** (0.020)	0.064*** (0.020)
Average SPI_{it}	-0.049 (0.288)							
$\text{Average SPI}_{it} > 1$		-0.012 (0.133)						
$\text{Average SPI}_{it} < 1$			0.179* (0.107)					
$\text{Flood frequency}_{it}$				0.125 (0.105)				

(continued)

Table 11. (continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Flood severity _{it}					0.116 (0.106)			
Flood magnitude _{it}						0.055 (0.043)		
Monsoon average pre 5Y _{it}							-6.3e-05 (0.000)	
Monsoon average pre 2Y _{it}								-1.4e-04** (0.000)
Origin-state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Destination-state/time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1860	1860	1860	1860	1860	1860	1860	1860
R ²	0.691	0.691	0.695	0.692	0.692	0.692	0.693	0.696

Note: The dependent variable is the bilateral migration rate from state *i* to state *j* between year *t* - 1 and year *t*. The subscript *t* indicates only that the variable varies over time. Robust standard errors in parentheses.

p* < 0.10, *p* < 0.05, ****p* < 0.01.

separately. Again, we find a statistically significant positive impact of drought, but the excess precipitation measure based on the continuous value of the SPI is negative and nonsignificant.

In Columns (4)–(6), we test three types of alternative flood measures based on the Dartmouth Flood Observatory data used in Ghimire and Ferreira (2016). The Observatory records every large flood observed (see definition in [Supplementary Appendix Table SA2](#)). In [Table 11](#), all the measures are based on floods occurring 1 year before migration, and we define the variables to be comparable with the previously used excess precipitation frequency measures. The flood frequency variable measures the number of months with a large flood event, flood severity measures the average flood severity index defined by the Observatory, and flood magnitude is the log of the product of frequency and duration. Although the signs of the flood variables are positive as expected *ex ante* (and in contrast to the SPI-based measures), none of the coefficients is statistically significant.

Columns (7) and (8) present estimations with yearly averaged precipitation for the summer monsoon months only (June through September). The estimation in Column (7) includes the average monsoon precipitation in the 5 years before migration (similar to the climate variability variables), and Column (8) uses the average during the 2 years preceding migration. We construct this measure to account for the rainy season only, since most yearly precipitation falls during the monsoon which is important for the agricultural sector. This avoids the smoothing of the monsoon impact by averaging over 12 months. We observe a significant effect only for the 2-year averaged monsoon, and more importantly, the effect is negative. This result suggests that the negative impact on migration of our measures of excess precipitation is not due to omission of the effects of monsoon periods and that precipitation data on their own do not capture all types of flood events in India.

5.3.2 Alternative specification

[Supplementary Appendix Table SD7](#) presents the same estimations as in [Table 2](#) but with an OLS rather than a PPML estimator and with standard errors clustered at the origin state. Clustering is not possible in PPML estimations, and can be an issue if there is spatial correlation in climatic factors between bordering states.¹⁹ The effect of drought and excess precipitation varies very little, presenting an even larger effect and of higher statistical significance, with the exception of drought magnitude which turns out to be nonsignificant compared to the results in [Table 2](#).

5.4 Discussion: Is drought-induced migration important?

The estimates of the impact of drought frequency and magnitude may seem small in relative terms. Two considerations are necessary before drawing policy implications. The first is that the effect is estimated only on inter-state migration, where migration barriers are high as shown in the estimation results, and the estimate is likely to represent a lower bound for internal migration in India, since intra-state migration represents the larger part of internal migration (see [Supplementary Appendix Table SC1](#)). The second point is that our estimates represent the effect on out-migration from one state to another specific state, and not the

19 The principal component analysis of state climatic factors shows that the state-level measures should be independent, though.

Table 12. Estimated drought-induced migration figures

1996–2001					2000–2001		
Migrants	Marginal drought effect	Minimum	Mean	Maximum	Minimum	Mean	Maximum
56,166,947	842,504	0	11,963,560	37,912,689	0	2,392,712	7,582,537

effect on total out-migration. Thus, the marginal effect obtained here is an underestimation of the effect on total internal migration. Table 12 presents a back-of-the-envelope calculation of what the estimates imply for migrant flows.

Table 12 shows the total flow of internal migrants between 1996 and 2001 in India. If we apply the estimated marginal effect of drought of 1.5% to total migrants, we obtain 842,504 migrants over 5 years for each additional month of drought. The summary statistics in Supplementary Appendix Table SA2 show that over 5 years, the number of droughts ranged from 0 and 45 months, with a mean of 14.2 months. We multiply each of the three numbers (minimum, mean, and maximum) by the marginal effect on the migrant population to see how many additional migrants were due to drought frequency in the best, mean, and worst-case scenarios. According to our estimate of the marginal effect of drought, in the mean scenario, over 5 years, there are 11.96 million additional migrants, and in the worst-case, there are 37.91 million migrants due to drought. These numbers correspond to 2.39 million and 7.58 million in 1 year. When converting the marginal effect into numbers of potentially drought-induced migrants, the effect is large.

These indicative numbers of past migrants due to drought frequency can be compared to Internal Displacement Monitoring Centre (IDMC) data which indicate that 3.7 million people were displaced by natural disasters in India in 2015. From the estimated effect for the earlier period 1996–2001, we obtain a yearly mean of 2.4 million displaced by drought alone. We emphasize that these figures are for illustrative purposes only, and are not projections of future migrations induced by drought. Compared to estimates from other neighboring countries, Hassani-Mahmooei and Parris (2012) predict between 3 million and 10 million internal migrants in Bangladesh over the next 40 years.

6. Conclusions

We analyze whether climate variability affects Indian bilateral inter-state migration rates using census data. The analysis in this article is one of the first attempts to investigate the impact of climate variability on internal migration using precise and complete census data at the level of a large and diverse country—India. Use of migration flow data defined between years $t-1$ and t allows us to test and compare the results from different timings of climatic factors prior to migration. This is a novelty compared to the existing literature on climatic factors and migration which mainly average figures over a longer time period. The other main contribution of the article is that we use objective meteorological indicators of climate variability based on the SPI. We created three variables based on the SPI: frequency, duration, and magnitude of drought and excess precipitation. In contrast to most previous studies, our analysis takes account of over-control bias that arises from including migration

determinants that are dependent on climatic factors such as average income. We explored separately three important channels through which climate variability could affect migration: average income, agricultural income, and urbanization. The use of census data allowed us to analyze the effect of climate variability on actual migration flows from rural areas, rather than using the urbanization rate as a proxy, as is frequent in the literature.

The estimation results show significant effects on bilateral migration rates from drought frequency, with an average effect of 1.5%. For agricultural states, the effect of drought frequency is 1.7% and bilateral migration rates also increase following an increase in the magnitude of drought in agricultural origin states. We suggest that the findings for drought frequency could be interpreted as evidence of migration induced by expectations of future droughts. Observed drought frequency tends to reinforce future expectations of drought, and hence, may induce migration. However, the relative effect is small compared to the important role of the barriers to migration in the Indian context, which explain the low Indian inter-state migration rates. It is possible that the impact of climate variability on internal migration in India is underestimated. When complete origin-destination district-level data become available, more detailed analysis of inter-district bilateral migration should be conducted. However, by extrapolating the estimated marginal effect on inter-state flows to total migration flows in India, we show that drought may have induced 2 million migrants on average in year 2001, a figure that is compatible with current IDMC estimates of displaced people in India following natural disasters.

Excess precipitation was not found to induce inter-state migration, contrary to what might have been expected *ex ante*. The only significant effect is that each additional month in a consecutive spell of excessive precipitation reduces bilateral migration rates by 1.5%. Alternative flood measures from the Dartmouth Flood Observatory are also never significant, although they show the expected positive effect on migration. An extended analysis using more exact flood measures, for example based on remote-sensing data as in [Guiteras et al. \(2015\)](#) or in [Gröger and Zylberberg \(2016\)](#), would be a fruitful direction for future research.

We found evidence of two possible channels through which climate variability affects inter-state migration in India: the impact on net state domestic product and the agricultural sector. In particular, the effect of drought frequency on rural out-migration is higher than on total inter-state migration, and the strongest impact is on rural–rural migration. However, a direct effect of drought frequency remains after controlling for these indirect drivers of migration. The analyses in this article could be extended when bilateral migration data from the 2011 Census become available. They could include analysis of potential insurance mechanisms via an institutional channel, in particular from the National Rural Employment Guarantee Act (NREGA) implemented in 2006.

Supplementary material

[Supplementary material](#) is available at *Cesifo* online.

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